

3.7 Ratio Test Worksheet

Determine whether each of the following series converge or diverge. You may use any theorem from 3.7 or earlier.

$$1. \sum_{k=1}^{\infty} \frac{k^3}{3^k}.$$

We note that the series will produce only non-zero terms.

$$\rho = \lim_{k \rightarrow \infty} \frac{|a_{k+1}|}{|a_k|} = \lim_{k \rightarrow \infty} \frac{\left(\frac{(k+1)^3}{3^{k+1}} \right)}{\left(\frac{k^3}{3^k} \right)} = \lim_{k \rightarrow \infty} \frac{(k+1)^3}{3^{k+1}} \cdot \frac{3^k}{k^3} = \lim_{k \rightarrow \infty} \frac{(k+1)^3}{3k^3} = \lim_{k \rightarrow \infty} \frac{1}{3} \left(\frac{k+1}{k} \right)^3 = \frac{1}{3} \left(\lim_{k \rightarrow \infty} \left(\frac{k+1}{k} \right) \right)^3 = \frac{1}{3} (1)^3 = \frac{1}{3} < 1.$$

Therefore, by the ratio test, the series $\sum_{k=1}^{\infty} \frac{k^3}{3^k}$ converges. ■

$$2. \sum_{k=1}^{\infty} \frac{2^{2k+1}}{3^k}$$

We can rewrite the series as $\sum_{k=1}^{\infty} \frac{2^{2k+1}}{3^k} = \sum_{k=1}^{\infty} \frac{2(4)^k}{3^k} = \sum_{k=1}^{\infty} \frac{8}{3} \cdot \left(\frac{4}{3} \right)^{k-1}$.

We note that this is a geometric series with $r = \frac{4}{3}$ and since $|r| > 1$ this series must diverge. ■

$$3. \sum_{k=1}^{\infty} \frac{(2k)!}{4^k}$$

We note that the series will produce only non-zero terms.

$$\begin{aligned} \rho &= \lim_{k \rightarrow \infty} \frac{|a_{k+1}|}{|a_k|} = \lim_{k \rightarrow \infty} \frac{\left(\frac{(2(k+1))!}{4^{k+1}} \right)}{\left(\frac{(2k)!}{4^k} \right)} = \lim_{k \rightarrow \infty} \frac{(2k+2)!}{4^{k+1}} \cdot \frac{4^k}{(2k)!} = \lim_{k \rightarrow \infty} \frac{(2k+2)!}{4(2k)!} = \lim_{k \rightarrow \infty} \frac{(2k+2)(2k+1)(2k)!}{4(2k)!} \\ &= \lim_{k \rightarrow \infty} \frac{(2k+2)(2k+1)}{4} = \infty \end{aligned}$$

Therefore, by the ratio test, the series $\sum_{k=1}^{\infty} \frac{(2k)!}{4^k}$ diverges. ■

4. Consider the series $\sum_{k=1}^{\infty} \frac{1}{2k-1}$.

i. What does the ratio test conclude about the series?

We note that the series will produce only non-zero terms.

$$\rho = \lim_{k \rightarrow \infty} \frac{|a_{k+1}|}{|a_k|} = \lim_{k \rightarrow \infty} \frac{\left(\frac{1}{2(k+1)-1}\right)}{\left(\frac{1}{2k-1}\right)} = \lim_{k \rightarrow \infty} \frac{1}{2k+1} \cdot \frac{2k-1}{1} = \lim_{k \rightarrow \infty} \frac{2k-1}{2k+1} = 1. \text{ So, the ratio test is inconclusive.}$$

ii. Use any other method to determine whether or not the given series diverges.

Let $f(x) = \frac{1}{2x-1}$. We note that $f(x)$ is a continuous, positive, and decreasing function on $[1, \infty)$,

therefore we will use the integral test. We investigate the integral $\int_1^{\infty} \frac{1}{2x-1} dx$.

$$\int_1^{\infty} \left(\frac{1}{2x-1}\right) dx = \lim_{t \rightarrow \infty} \left(\int_1^t \left(\frac{1}{2x-1}\right) dx \right) = \lim_{t \rightarrow \infty} \left(\left(\frac{1}{2} \ln |2x-1| \right) \Big|_1^t \right) = \lim_{t \rightarrow \infty} \left(\frac{1}{2} \ln |2t-1| - \frac{1}{2} \ln |1| \right) = \infty$$

Therefore, by the integral test, since $\int_1^{\infty} \frac{1}{2x-1} dx$ diverges the series $\sum_{k=1}^{\infty} \frac{1}{2k-1}$ diverges.

OR

We first see that $\sum_{k=1}^{\infty} \frac{1}{2k-1} = \frac{1}{2} + \sum_{k=2}^{\infty} \frac{1}{2k-1}$. We then consider the series $\sum_{k=2}^{\infty} \frac{1}{k-1}$ which we note is the

Harmonic series and therefore diverges. We note that both of these series contain only positive terms. We calculate:

$$\lim_{k \rightarrow \infty} \frac{\left(\frac{1}{2k-1}\right)}{\left(\frac{1}{k-1}\right)} = \lim_{k \rightarrow \infty} \frac{k-1}{2k-1} = \lim_{k \rightarrow \infty} \frac{k-1}{2k-1} \cdot \frac{1/k}{1/k} = \lim_{k \rightarrow \infty} \frac{1-1/k}{2-1/k} = \frac{1}{2}$$

Since we obtained a positive and finite result we know, by the Limit Comparison Test, that both series have the same behavior. Therefore, since $\sum_{k=2}^{\infty} \frac{1}{k-1}$ converges then $\sum_{k=2}^{\infty} \frac{1}{2k-1}$ diverges and so

$$\sum_{k=1}^{\infty} \frac{1}{2k-1} = \frac{1}{2} + \sum_{k=2}^{\infty} \frac{1}{2k-1} \text{ must also diverge.}$$

5. $\sum_{k=1}^{\infty} \frac{2 + (-1)^k}{5^k}$.

Here the ratio test here will be inconclusive (details not shown). We will progress by using the comparison theorem. We note that $\frac{2 + (-1)^k}{5^k} \leq \frac{3}{5^k}$ for $k \geq 1$. The series

$$\sum_{k=1}^{\infty} \frac{3}{5^k} = \sum_{k=1}^{\infty} 3 \left(\frac{1}{5} \right)^k = \sum_{k=1}^{\infty} \frac{3}{5} \left(\frac{1}{5} \right)^{k-1} \text{ is a geometric series with } |r| = \left| \frac{1}{5} \right| < 1. \text{ Thus } \sum_{k=1}^{\infty} \frac{3}{5^k} = \frac{3/5}{1 - 1/5} = \frac{3}{5} \cdot \frac{5}{4} = \frac{3}{4}.$$

Therefore, by the comparison theorem, since $\sum_{k=1}^{\infty} \frac{3}{5^k}$ converges the series $\sum_{k=1}^{\infty} \frac{2 + (-1)^k}{5^k}$ converges. ■

6. $\sum_{k=1}^{\infty} \frac{k}{2^k}$

We note that the series contains only non-zero terms.

$$\text{We then calculate that } \lim_{k \rightarrow \infty} \frac{\left(\frac{k+1}{2^{k+1}} \right)}{\left(\frac{k}{2^k} \right)} = \lim_{k \rightarrow \infty} \frac{k+1}{2^{k+1}} \cdot \frac{2^k}{k} = \lim_{k \rightarrow \infty} \frac{k+1}{2k} = \lim_{k \rightarrow \infty} \frac{k+1}{2k} \cdot \frac{1/k}{1/k} = \lim_{k \rightarrow \infty} \frac{1+1/k}{2} = \frac{1}{2} < 1.$$

Therefore, the series converges by the Ratio Test. ■

7. $\sum_{k=1}^{\infty} \frac{(-1)^k \cdot k}{k^2 + 1}$

We note that this is an alternating series with $a_k = \frac{k}{k^2 + 1}$. Clearly

$$\frac{a_{k+1}}{a_k} = \frac{\left(\frac{k+1}{(k+1)^2 + 1} \right)}{\left(\frac{k}{k^2 + 1} \right)} = \frac{k+1}{k^2 + 2k + 2} \cdot \frac{k^2 + 1}{k} = \frac{k^3 + k^2 + k + 1}{k^3 + 2k^2 + 2k} = \frac{(k^3 + k^2 + k) + 1}{(k^3 + k^2 + k) + k^2 + k} < 1 \text{ and so this}$$

sequence is decreasing.

$$\text{Also } \lim_{k \rightarrow \infty} \frac{k}{k^2 + 1} = \lim_{k \rightarrow \infty} \frac{k}{k^2 + 1} \cdot \frac{1/k^2}{1/k^2} = \lim_{k \rightarrow \infty} \frac{1/k}{1 + 1/k^2} = \frac{0}{1 + 0} = 0.$$

Therefore, by the AST, this series converges. ■

8. $\sum_{k=1}^{\infty} \frac{k^2 + 2}{k^4}$

We first rewrite the series as $\sum_{k=1}^{\infty} \frac{k^2 + 2}{k^4} = \sum_{k=1}^{\infty} \left(\frac{1}{k^2} + \frac{2}{k^4} \right)$. We want to split this series into two but we can only do this if both resulting series are convergent.

We see that $\sum_{k=1}^{\infty} \frac{1}{k^2}$ is a p-series with $p = 2 > 1$ and so this series converges.

We see that $\sum_{k=1}^{\infty} \frac{2}{k^4} = 2 \sum_{k=1}^{\infty} \frac{1}{k^4}$ and this is a p-series with $p = 4 > 1$ and so this series converges.

Therefore we can say that $\sum_{k=1}^{\infty} \frac{k^2 + 2}{k^4} = \sum_{k=1}^{\infty} \left(\frac{1}{k^2} + \frac{2}{k^4} \right) = \sum_{k=1}^{\infty} \frac{1}{k^2} + \sum_{k=1}^{\infty} \frac{2}{k^4}$ and so this series converges. ■

9. Classify each of the following series as absolutely convergent, conditionally convergent, or divergent.

a) $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^2 + 3}$

We will use the Ratio test to check to see if the given series is absolutely convergent. We note that the series will produce only non-zero terms.

$$\text{We note that } \rho = \lim_{k \rightarrow \infty} \frac{\left| \frac{(-1)^{k+2}}{(k+1)^2 + 3} \right|}{\left| \frac{(-1)^{k+1}}{k^2 + 3} \right|} = \lim_{k \rightarrow \infty} \frac{1}{(k+1)^2 + 3} \cdot \frac{k^2 + 3}{1} = \lim_{k \rightarrow \infty} \frac{k^2 + 3}{k^2 + 2k + 4} = 1, \text{ so the Ratio test is}$$

inconclusive.

We instead note that $\sum_{k=1}^{\infty} \left| \frac{(-1)^{k+1}}{k^2 + 3} \right| = \sum_{k=1}^{\infty} \frac{1}{k^2 + 3}$ and that $\frac{1}{k^2 + 3} < \frac{1}{k^2}$ for all k . Since $\sum_{k=1}^{\infty} \frac{1}{k^2}$ is a convergent p -

series with $p = 2 > 1$ we know, by the Comparison test, that $\sum_{k=1}^{\infty} \frac{1}{k^2 + 3}$ must also converge. So the series

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^2 + 3} \text{ is absolutely convergent.}$$
■

$$b) \sum_{n=1}^{\infty} \frac{(-1)^{n+1} (n!)^2}{(2n)!}$$

We first check to see if the given series is absolutely convergent. We note that $\sum_{n=1}^{\infty} \left| \frac{(-1)^{n+1} (n!)^2}{(2n)!} \right| = \sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!}$.

We note that the series will produce only non-zero terms. Now, using the ratio test we see that:

$$\rho = \lim_{n \rightarrow \infty} \frac{\frac{((n+1)!)^2}{(2(n+1))!}}{\frac{(n!)^2}{(2n)!}} = \lim_{n \rightarrow \infty} \frac{(n+1)! \cdot (n+1)!}{(2n+2)!} \cdot \frac{(2n)!}{n! \cdot n!} = \lim_{n \rightarrow \infty} \frac{(n+1)(n+1)}{(2n+2)(2n+1)} = \lim_{n \rightarrow \infty} \frac{n^2 + 2n + 1}{4n^2 + 6n + 2} = \frac{1}{4} < 1. \text{ So, by the}$$

ratio test $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} (n!)^2}{(2n)!}$ is absolutely convergent.

■